Recall — speed, distance and time

One formula, three arrangements, and the units that keep them honest.

LESSON 116 | 1 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 12 RATES

LESSON CLOCK

10' | 12' | 12' | 6'

The formula	10 min
Rearranging	12 min
Units	12 min
Converting	6 min

LEARNING OBJECTIVES

- State the relation between speed, distance and time.
- Rearrange it to find any one of the three.
- Use consistent units.
- Convert between *km/h* and *m/s*.

TEXTBOOK

National	pp. 335–338
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.4

01 Explanation

The formula

$$s = vt$$

Distance equals speed times time — because a speed of *60* km/h means *60* km covered in each hour.

SPEED	TIME	DISTANCE
60 km/h	*3 h*	*180 km*
80 km/h	*2.5 h*	*200 km*
5 m/s	*12 s*	*60 m*
4 km/h	*45 min*	*3 km*

THE LAST ROW NEEDS THE TIME IN HOURS

45 minutes is 0.75 h, and $4 \cdot 0.75 = 3$ km. Using *45* with a speed in km/h gives *180* km, which is absurd — and the absurdity is the check.

Rearranging

$s = vt$ $v = \frac{s}{t}$ $t = \frac{s}{v}$

WANTED	GIVEN	WORKING	ANSWER
distance	$v = 60, t = 3$	$60 \cdot 3$	180 km
speed	$s = 180, t = 3$	$180 \div 3$	60 km/h
time	$s = 180, v = 60$	$180 \div 60$	3 h

ONE FORMULA, NOT THREE

$s = vt$ is all that has to be remembered; the other two are it rearranged. Writing the one you know and rearranging is safer than memorising a triangle.

Units

SPEED IN	DISTANCE IN	TIME IN
km/h	km	hours
m/s	m	seconds
m/min	m	minutes

TIME GIVEN	IN HOURS
30 min	0.5
45 min	0.75
20 min	$\frac{1}{3}$
1 h 30 min	1.5

THE UNITS OF THE ANSWER ARE BUILT FROM THE OTHERS

Kilometres divided by hours gives km/h automatically. Writing the units through the calculation makes the answer's unit appear by itself.

Converting

$$1\ km/h = \frac{1000}{3600}\ m/s = \frac{5}{18}\ m/s$$

KM/H	M/S
36	10
72	20
18	5
90	25

DIVIDE BY 3.6 TO GO FROM KM/H TO M/S

And multiply by 3.6 to come back. A sprinter at 10 m/s is running at 36 km/h, which is a useful pair of numbers to remember.

02 Worked examples

EXAMPLE 1 **A car travels at $80\ km/h$ for 2.5 hours. How far does it go?**

1. $s = vt$
2. $= 80 \cdot 2.5$
3. $= 200$ km.

ANSWER

$200\ km$

EXAMPLE 2 **A cyclist covers *3 km* in *45* minutes. Find the speed in km/h.**

1. 45 min = 0.75 h.
   Convert first.
2. $v = \dfrac{3}{0.75}$
3. $= 4$ km/h.

ANSWER

4 km/h

EXAMPLE 3 **Convert *72 km/h* to metres per second.**

1. $72 \div 3.6$
2. $= 20$ m/s.
3. Check: $20 \cdot 3.6 = 72$ ✓

ANSWER

20 m/s

03 Interactive model

Time a pupil walking a measured 20 m and compute the speed in both units; the numbers are their own and the units matter immediately.

Which of the three is wanted?

$v = 60$, $t = 3$: the distance is:

- 20
- 63
- 180
- 180 km

$s = 180$, $t = 3$: the speed is:

- 540
- 60 km/h
- 60 km
- 3

$s = 180$, $v = 60$: the time is:

- 3 h
- 120
- 240
- 0.33

45 minutes in hours:

- 0.45
- 0.75
- 4.5
- 45

20 minutes in hours:

- 0.2
- $\frac{1}{3}$
- 0.5
- 2

36 km/h in m/s:

- 10
- 36
- 100
- 3.6

20 m/s in km/h:

- *36*
- *72*
- *60*
- *7.2*

Using minutes with a speed in km/h gives:

- the right answer
- an absurd answer
- metres
- nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Speed	Tezlik	Скорость
Distance	Masofa	Расстояние
Time	Vaqt	Время
Kilometres per hour	km/soat	км/ч
Metres per second	m/sek	м/с
To rearrange	O‘zgartirib yozish	Выразить
Consistent units	Mos birliklar	Согласованные единицы
Rate	Tezlik (me’yor)	Норма

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

s equals:

- vt
- $\frac{v}{t}$
- $\frac{t}{v}$
- $v + t$

v equals:

- st
- $\frac{s}{t}$
- $\frac{t}{s}$
- $s - t$

t equals:

- sv
- $\frac{v}{s}$
- $\frac{s}{v}$
- $s + v$

With a speed in km/h, the time must be in:

- minutes
- hours
- seconds
- any unit

72 km/h in m/s is:

- 20
- 26
- 7.2
- 259

A sprinter at 10 m/s runs at:

- 10 km/h
- 36 km/h
- 60 km/h
- 3.6 km/h

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $v = 60$, $t = 3$: the distance

   ANSWER *180 km*

2. $v = 80$, $t = 2.5$

   ANSWER *200 km*

3. $v = 5$ m/s, $t = 12$ s

   ANSWER *60 m*

4. $s = 180$, $t = 3$: the speed

   ANSWER *60 km/h*

5. $s = 180$, $v = 60$: the time

   ANSWER *3 h*

6. 30 min in hours

   ANSWER *0.5*

7. 45 min in hours

   ANSWER *0.75*

Medium · the standard question

7 PROBLEMS

1. 3 km in 45 min: the speed

   ANSWER 4 km/h

2. 36 km/h in m/s

   ANSWER 10

3. 72 km/h in m/s

   ANSWER 20

4. 25 m/s in km/h

   ANSWER 90

5. $v = 4$ km/h, $t = 45$ min

   ANSWER 3 km

6. $s = 150$ km, $v = 60$ km/h

   ANSWER 2.5 h

7. 20 min in hours

   ANSWER $\frac{1}{3}$

Hard · stretch and reason

7 PROBLEMS

1. A train covers 210 km in $2\ h\ 30\ min$

   ANSWER $84\ km/h$

2. A walker at 5 km/h covers 4 km in

   ANSWER 48 min

3. A runner at 8 m/s in 90 s

   ANSWER $720\ m$

4. Convert 54 km/h to m/s

   ANSWER 15

5. Convert 12 m/s to km/h

   ANSWER 43.2

6. A car at 90 km/h covers 1 km in

   ANSWER 40 s

7. Why must units match?

   ANSWER The formula divides one by the other, so they must belong together

07 Homework

Homework — 5 tasks

Convert the time to hours before using a speed in km/h.

1. A bus travels at 70 km/h for 4 hours. Find the distance.
2. A car covers 240 km in 3 hours. Find its speed.
3. How long does 150 km take at 50 km/h?
4. A cyclist covers 9 km in 40 minutes. Find the speed in km/h.
5. Convert 108 km/h to m/s and 15 m/s to km/h.